US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12407805
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Mori; Kensaku et al.

Image capturing apparatus for capturing a plurality of eyeball images, image capturing method for image capturing apparatus, and storage medium

Abstract

An image capturing apparatus includes a first line-of-sight detector configured to capture a first eyeball image as a left eyeball image and detect a first line of sight based on the first eyeball image at a first cycle, and a second line-of-sight detector configured to capture a second eyeball image as a right eyeball image and detect a second line of sight based on the second eyeball image at the first cycle. A timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image are different from each other.

Inventors: **Mori; Kensaku (Kanagawa, JP), Mine; Yosuke (Tokyo, JP)**

Applicant: **CANON KABUSHIKI KAISHA** (Tokyo, JP)

Family ID: **89576218**

Assignee: **Canon Kabushiki Kaisha (Tokyo, JP)**

Appl. No.: **18/355152**

Filed: **July 19, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240031551 A1	Jan. 25, 2024

Foreign Application Priority Data

JP	2022-116420	Jul. 21, 2022

Publication Classification

Int. Cl.: **H04N13/296** (20180101); **G06F3/01** (20060101); **H04N13/239** (20180101)

U.S. Cl.:

CPC **H04N13/296** (20180501); **G06F3/013** (20130101); **H04N13/239** (20180501);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2011/0249868	12/2010	Tsukizawa	382/103	A61B 5/163
2019/0232501	12/2018	Funazukuri	N/A	G06N 3/008
2021/0357027	12/2020	Yamasaki	N/A	H04N 23/60
2021/0389595	12/2020	Yasuda	N/A	G06F 3/04842
2022/0003632	12/2021	Glasenapp	N/A	G02C 7/027
2022/0036779	12/2021	Nishibe	N/A	G06T 7/20
2022/0257110	12/2021	Ito	N/A	A61B 3/032
2023/0224561	12/2022	Ito	348/262	G06V 40/18
2023/0393655	12/2022	Mizuo	N/A	G02B 27/0093

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H11259225	12/1998	JP	N/A
2002007053	12/2001	JP	N/A
2019063310	12/2018	JP	N/A
2022038815	12/2021	WO	N/A

Primary Examiner: Nawaz; Talha M

Attorney, Agent or Firm: Canon U.S.A., Inc., IP Division

Background/Summary

BACKGROUND

Field

(1) The present disclosure relates to an image capturing apparatus for capturing a plurality of eyeball images, an image capturing method for the image capturing apparatus, and a storage medium.

Description of the Related Art

(2) There is a mixed reality (MR) technique called “X Reality” or “Extended Reality”. The MR technique combines physical and virtual spaces seamlessly. With an MR system using a video-seethrough head-mounted display (HMD), a user wearing the HMD sees a composite image created by superimposing a computer graphics (CG) image on a physical space image captured by an image capturing unit included in the HMD. These images each correspond to the right and left eyes

separately, presenting a three-dimensional MR space in a stereo moving image to a user wearing an HMD. An MR space less likely to cause a sense of awkwardness to a user can be provided by a technique, in addition to the above described technique, of acquiring eyeball images of a user wearing an HMD, detecting the directions of the lines of sight of the user wearing the HMD, and generating CG images suited to the lines of sight of the user wearing the HMD, by built-in units of the HMD. In recent years, there has been a demand for an enhanced performance of acquiring results of line-of-sight direction detections supporting improved frame rates of composite images to be provided to a user wearing an HMD in order to provide a higher quality MR space.
(3) There are known several techniques to meet the demand. For example, Japanese Patent Application Laid-Open No. 2019-63310 discloses a method for improving the accuracy of detecting lines of sight by once detecting positions of lines of sight of the left and right eyes of a user looking at a display screen for work, and combining and compositing results of the detected positions. Further, Japanese Patent Application Laid-Open No. 2002-7053 discloses a method for using a result of a determination that indicates that a user has blinked either his or her left eye or right eye without blinking the other in detection of blinking left and right eyes as a determination signal for performing input operations on a device.
(4) To provide a higher quality MR space, however, there is also a demand for speeding up a cycle of reading out the result through rapid acquisition of a line-of-sight direction detection result, in addition to the accuracy of line-of-sight detections. Japanese Patent Application Laid-Open No. 2019-63310 discusses a method for improving the accuracy of line-of-sight detections using a plurality of line-of-sight detecting units, but does not touch upon a method for improving a cycle of reading out a line-of-sight direction detection result. Japanese Patent Application Laid-Open No. 2002-7053 discloses a technique of determining a change in eye using a plurality of line-of-sight detecting units, but does not touch upon a method for improving a cycle of reading out a line-of-sight direction detection result.
(5) On the other hand, for the purpose of improving a cycle of reading out a line-of-sight direction detection result, implementing a high frame rate image capturing element and increasing communication bus width to improve data transfer bandwidth between devices leads to an increase in cost.

SUMMARY

(6) According to an aspect of the present disclosure, an image capturing apparatus includes a first line-of-sight detector configured to capture a first eyeball image as a left eyeball image and detect a first line of sight based on the first eyeball image at a first cycle, and a second line-of-sight detector configured to capture a second eyeball image as a right eyeball image and detect a second line of sight based on the second eyeball image at the first cycle. A timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image are different from each other.
(7) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a block diagram illustrating an example of an image capturing system.
(2) FIG. **1**B illustrates an example of an appearance of the image capturing system and a user wearing an image capturing apparatus.
(3) FIG. **2** illustrates an example of the configuration of a line-of-sight detection unit.
(4) FIG. **3** is a flowchart illustrating line-of-sight detection processing in the image capturing apparatus.

(5) FIG. **4** is a time chart illustrating an example of a control operation of the image capturing apparatus.
(6) FIG. **5** is a time chart illustrating an example of another control operation of the image capturing apparatus.
DESCRIPTION OF THE EMBODIMENTS
(7) Some exemplary embodiments will be described below with reference to the accompanied drawings. The following exemplary embodiments are not intended to limit the scope of the claimed disclosure, and not all combinations of the features described in the exemplary embodiments are used.
(8) FIGS. **1**A and **1**B are examples of the configuration of an image capturing system **1** according to a first exemplary embodiment. FIG. **1**A is a block diagram illustrating a configuration of the image capturing system **1**, and FIG. **1**B illustrates an example of an appearance of the image capturing system **1** and a user wearing an image capturing apparatus **10**. The image capturing system **1** includes the image capturing apparatus **10** as a head-mounted device (HMD) (hereinafter, referred to as HMD **10**), an image processing apparatus **30** (personal computer work station (PCWS)), and a cable **20**. The image processing apparatus **30** generates an image of a mixed reality (MR) space created by combining a physical space and a virtual space, and provides the image (hereinafter, referred to as composite image) to the HMD **10**. The cable **20** connects the HMD **10** and the image processing apparatus **30** to each other. A communication path is not limited to the cable **20** in wired connection, and a wireless connection communication path may be used.
(9) A configuration of the HMD **10** will now be described. The HMD **10**, which is an image capturing apparatus, includes a display element **110***a* for left eyes, a display element **110***b* for right eyes, a position-orientation sensor **120**, a background image capturing element **130***a* for left eyes, a background image capturing element **130***b* for right eyes. The HMD **10** further includes delaying units **132***a* and **132***b*, a position detection image capturing element **140***a* for left eyes, a position detection image capturing element **140***b* for right eyes, a line-of-sight detection unit (detector) **150***a* for left eyes, and a line-of-sight detection unit (detector) **150***b* for right eyes.
(10) The image capturing elements **130***a*, **130***b*, **140***a*, and **140***b* are complementary metal oxide semiconductors (CMOS), charge coupled devices (CCD), or another type of device. The image capturing elements **130***a*, **130***b*, **140***a*, and **140***b* acquire captured images of a physical space via an optical system in accordance with set exposure times, sensor gains, exposure start timings, and so on, based on control signals from a control unit **100**.
(11) The display elements **110***a* and **110***b* are organic light-emitting diodes (OLED) displays, or liquid crystal displays, or another type of device. The display elements **110***a* and **110***b* present composite images through an optical system to a person wearing the HMD **10** (user).
(12) The image capturing system **1** deals with stereoscopic images using images for a right eye and images for a left eye. For this reason, the HMD **10** includes the background image capturing element **130***a* for left eyes, the background image capturing element **130***b* for right eyes, the position detection image capturing element **140***a* for left eyes, and the position detection image capturing element **140***b* for right eyes. The HMD **10** further includes the display element **110***a* for left eyes, the display element **110***b* for right eyes, the line-of-sight detection unit **150***a* for left eyes, and the line-of-sight detection unit **150***b* for right eyes.
(13) The background image capturing elements **130***a* and **130***b* capture background images that serve as a basis for a composite image. The position detection image capturing elements **140***a* and **140***b* capture images for position detection for generating computer graphics (CG) images. Background images and position detection images are different from each other in angle of view, resolution, image processing, and the like, so that different image capturing elements, i.e., the background image capturing elements **130***a* and **130***b*, and the position detection image capturing elements **140***a* and **140***b*, are used. Background images and position detection images may be generated by performing cut-out from captured images using a type of element common to the

background image capturing elements and the position detection image capturing elements.
(14) The position-orientation sensor **120** detects position-orientation information about the HMD **10**. The delaying units **132***a* and **132***b* delay display vertical synchronization signals input from the control unit **100**, generate line-of-sight detection start signals, and output the line-of-sight detection start signals to line-of-sight detection units **150***a* and **150***b*, respectively. The line-of-sight detection unit **150***a* for left eyes performs line-of-sight detection of the line of sight of the left eye of a user wearing the HMD **10** based on a line-of-sight detection start signal, and outputs a result of the detection to the control unit **100**. The line-of-sight detection unit **150***b* for right eyes performs line-of-sight detection of the line of sight of the right eye of the user wearing the HMD **10** based on a line-of-sight detection start signal, and outputs a result of the detection to the control unit **100**.
(15) The control unit **100** performs image processing on captured images and display images, and controls the components described above to operate in collaboration with the image processing apparatus **30**. Known techniques are used for capturing and displaying stereoscopic images, and a description thereof will be omitted.
(16) The image processing apparatus **30** includes a position detecting unit **300**, a CG generating unit **310**, and a compositing unit **320**. The position detecting unit **300** receives position detection images transmitted from the HMD **10**, position-orientation information about the HMD **10**, and display positions of the display elements **110***a* and **110***b* as line-of-sight detection results from the line-of-sight detection units **150***a* and **150***b*. The position detecting unit **300** outputs information about position, size, orientation, and/or the like for generating a CG image of a virtual space to be presented to a user wearing the HMD **10** based on the received pieces of information.
(17) The CG generating unit **310** generates a predetermined CG image for the left eye and a predetermined CG image for the right eye based on the output from the position detecting unit **300**. The CG images go through rendering processing based on computer-aided design (CAD) data stored in a hard disk drive (HDD) or another type of storage medium in the image processing apparatus **30**
(18) The compositing unit **320** superimposes CG images on background images transmitted from the HMD **10**, creates a composite image, and outputs the composite image as a display image to the HMD **10**.
(19) Next, the delaying units **132***a* and **132***b* will be described. The delaying units **132***a* and **132***b* receive display vertical synchronization signals that each indicate the beginning of each frame of a composite image via the control unit **100** from the compositing unit **320**. The display vertical synchronization signals are input to the delaying units **132***a* and **132***b*. The delaying units **132***a* and **132***b* delay the display vertical synchronization signals by delay amounts set by the control unit **100** beforehand, generate line-of-sight detection start signals, and output the line-of-sight detection start signals to the line-of-sight detection units **150***a* and **150***b*, respectively.
(20) FIG. **2** illustrates an example of a configuration of the line-of-sight detection unit **150***a* of FIG. **1**A. A line-of-sight detection start signal from the delaying unit **132***a* is input to the line-of-sight detection unit **150***a*. The line-of-sight detection unit **150***a* then performs line-of-sight detection, and outputs the result of the detection to the control unit **100**.
(21) Units of the line-of-sight detection unit **150***a* will be described. The line-of-sight detection unit **150***a* includes a line-of-sight central processing unit (CPU) **3**, a memory unit **4**, an analog/digital (A/D) conversion circuit **201**, an illumination light source driving circuit **205**, an eyeball image capturing element **17**, and illumination light sources **13***a* to **13***f*. The line-of-sight detection CPU **3** is a central processing unit of a microcomputer. The line-of-sight detection CPU **3** is connected to the A/D conversion circuit **201**, the illumination light source driving circuit **205**, and the memory unit **4**.
(22) The eyeball image capturing element **17** captures eyeball images of the left eye of a user wearing the HMD **10**. The A/D conversion circuit **201** performs A/D conversion on each eyeball image captured by the eyeball image capturing element **17**, and outputs a digital eyeball image to

the line-of-sight detection CPU **3**. The line-of-sight detection CPU **3** writes the digital eyeball image in the memory unit **4**.
(23) When the line-of-sight detection start signal from delaying unit **132***a* is input, the line-of-sight detection CPU **3** controls the A/D conversion circuit **201** and the illumination light source driving circuit **205** to start line-of-sight detection processing. The line-of-sight detection CPU **3** acquires the digital eyeball image from the A/D conversion circuit **201**, extracts features of the digital eyeball image for line-of-sight detection following a predetermined algorithm, and calculates a line-of-sight of the left eye of the user wearing the HMD **10** from positions of the features. The line of sight is calculated as coordinates of a composite image to be displayed by the display element **110***a*. The line-of-sight detection CPU **3** sends the coordinates of the line of sight as the calculated line-of-sight detection result to the control unit **100**.
(24) The illumination light sources **13***a* to **13***f* emit light to the eyeball of the user wearing the HMD **10** to detect the direction of the line of sight from a relationship between the pupil and an image created by the cornea's reflection of light emitted from light sources in a single-lens reflex camera or another device. The illumination light sources **13***a* to **13***f* include infrared light emitting diodes, and are arranged around a lens unit connected to the eyeball image capturing element **17**.
(25) The image of the illuminated eye and the image created by the cornea reflection with the illumination light sources **13***a* to **13***f* are formed on the eyeball image capturing element **17** formed in a two-dimensional array of photoelectronic elements, such as CMOSs, by the lens unit. The lens unit positions the pupil of the eye of the user wearing the HMD **10** and the eyeball image capturing element **17** in a conjugate relationship.
(26) The line-of-sight detection CPU **3** uses the predetermined algorithm to detect the line of sight from the positional relationship between the eyeball as the image formed on the eyeball image capturing element **17** and the image created by the cornea's reflection with the illumination light sources **13***a* to **13***f*. Specifically, the line-of-sight detection CPU **3** detects the pupil's center and the image created by the cornea's reflection with the illumination light sources **13***a* to **13***f* from the eyeball image captured on the eyeball image capturing element **17** while the eyeball is illuminated by light emitted by the illumination light sources **13***a* to **13***f* in the predetermined line-of-sight detection algorithm. The line-of-sight detection CPU **3** calculates an angle of rotation of the eyeball from the image created by the cornea's reflection and the pupil's center, and determines the line of sight of the user wearing the HMD **10** from the angle of rotation.
(27) The line-of-sight detection unit **150***b* has the same functions as the line-of-sight detection unit **150***b*, as well as the internal configuration. The line-of-sight detection unit **150***b* detects the line of sight of the right eye of the user wearing the HMD **10**. The line of sight is calculated as coordinates of a composite image to be displayed by the display element **110***b*.
(28) FIG. **3** is a flowchart illustrating a method for image capture by the HMD **10**. A method will be described that improves a cycle of reading out a line-of-sight detection result by using the line-of-sight detection units **150***a* and **150***b* and controlling line-of-sight detection start signals to be transmitted to the line-of-sight detection units **150***a* and **150***b*, respectively, in an optimal manner.
(29) In step S**101**, when a power source button is pressed, the units in the image capturing system **1** are energized.
(30) In step S**102**, the control unit **100** sets initial delay amounts to the delaying unit **132***a* and the delaying unit **132***b*, respectively. Delay amounts different between the delaying units **132***a* and **132***b* are settable.
(31) In step S**103**, the control unit **100** determines whether a display vertical synchronization signal indicating the beginning of each frame of a composite image is received from the compositing unit **320** in the image processing apparatus **30**. If no display vertical synchronization signal is received (NO in step S**103**), the processing returns to step S**103**. If a display vertical synchronization signal is received (YES in step S**103**), the processing proceeds to step S**104**.
(32) In step S**104**, the control unit **100** outputs a display vertical synchronization signal to the

delaying unit **132***a* and a display vertical synchronization signal to the delaying unit **132***b*, respectively. After the display vertical synchronization signal is input to the delaying unit **132***a* and a period corresponding to the delay amount elapses, the delaying unit **132***a* outputs a line-of-sight detection start signal to the line-of-sight detection unit **150***a*. After the display vertical synchronization signal is input to the delaying unit **132***b* and a period corresponding to the delay amount elapses, the delaying unit **132***b* outputs a line-of-sight detection start signal to the line-of-sight detection unit **150***b*.
(33) In step S**105**, on receiving the line-of-sight detection start signal, the line-of-sight detection unit **150***a* performs line-of-sight detection processing on the left eye of the user wearing the HMD **10** to detect the line of sight of the left eye of the user wearing the HMD **10**. On receiving the line-of-sight detection start signal, the line-of-sight detection unit **150***b* performs line-of-sight detection processing on the right eye of the user wearing the HMD **10** to detect the line of sight of the right eye of the user wearing the HMD **10**.
(34) In step S**106**, the line-of-sight detection unit **150***a* outputs the line-of-sight detection result of the left eye of the user wearing the HMD **10** to the control unit **100**, and the line-of-sight detection unit **150***b* outputs the line-of-sight detection result of the right eye of the user wearing the HMD **10** to the control unit **100**. After that, the processing returns to step S**103**.
(35) FIG. **4** is a time chart illustrating an example of a control operation performed by the HMD **10** and the relationship of control signals between units. The left side of FIG. **4** illustrates timings when the compositing unit **320** transmits display vertical synchronization signals via the control unit **100** to the delaying units **132***a* and **132***b*. The right side of FIG. **4** illustrates timings when the line-of-sight detection units **150***a* and **150***b* output line-of-sight detection results to the control unit **100**.
(36) The control unit **100** receives a display vertical synchronization signal **115** from the compositing unit **320** at a frame cycle of T. After that, the control unit **100** outputs a display vertical synchronization signal **115***a* to the delaying unit **132***a* and a display vertical synchronization signal **115***b* to the delaying unit **132***b*. It is assumed herein that the display vertical synchronization signals **115***a* and **115***b* are not delayed to each other.
(37) For example, the display vertical synchronization signal **115**, which indicates the beginning of each frame of a composite image, is output at a frame rate of 60 [Hz]. In this case, the inverse of the frame rate is approximately 16.67 [ms], which is the frame cycle T. The display vertical synchronization signal **115***a* is input to the delaying unit **132***a* and the display vertical synchronization signal **115***b* to the delaying unit **132***b* at a cycle of 16.67 [ms].
(38) Next, after the display vertical synchronization signal **115***a* is input to the delaying unit **132***a* and a period corresponding to a delay amount D**1** set by the control unit **100** elapses, the delaying unit **132***a* outputs a line-of-sight detection start signal **135** to the line-of-sight detection unit **150***a*. After the display vertical synchronization signal **115***b* is input to the delaying unit **132***b* and a period corresponding to a delay amount D**2** set by the control unit **100** elapses, the delaying unit **132***b* outputs a line-of-sight detection start signal **145** to the line-of-sight detection unit **150***b*.
(39) Here, the control unit **100** can set the delay amount D**2** based on the frame cycle T and the delay amount D**1** using the following expression:
$D2=(T-D1)/2$
(40) As described above, both of the line-of-sight detection units **150***a* and **150***b* can start line-of-sight detection at a cycle of T/2. In other words, a line-of-sight detection result **147** from the line-of-sight detection unit **150***a* and a line-of-sight detection result **148** from the line-of-sight detection unit **150***b* can be input to the control unit **100** at a cycle of T/2.
(41) For example, when the delay amount D**1** is 1 [ms], the delay amount D**2** is set to 9.3 [ms] as calculated as follows:
$D2=(16.67-1)/2=9.3$ [ms]
(42) Thus, 1 [ms] as D**1** after the display vertical synchronization signal **115** is input to the control

unit **100**, the line-of-sight detection unit **150***a* starts line-of-sight detection. Similarly, 9.3 [ms] as D**2** after the display vertical synchronization signal **115** is input to the control unit **100**, the line-of-sight detection unit **150***b* starts line-of-sight detection.

(43) After the line-of-sight detection start signal **135** is input to the line-of-sight detection unit **150***a*, the line-of-sight detection unit **150***a* starts line-of-sight detection processing on the left eye, and after a line-of-sight detection processing time **136** elapses, the line-of-sight detection unit **150***a* outputs the line-of-sight detection result **147** of the left eye to the control unit **100**. The line-of-sight detection processing time **136** is a time period from when the line-of-sight detection start signal **135** is input to the line-of-sight detection unit **150***a* to when the line-of-sight detection unit **150***a* outputs the line-of-sight detection result **147**. The line-of-sight detection processing includes performing exposure with the eyeball image capturing element **17**, reading out a result of the exposure with the eyeball image capturing element **17**, and performing line-of-sight detection processing using an eyeball image.

(44) After the line-of-sight detection start signal **145** is input to the line-of-sight detection unit **150***b*, the line-of-sight detection unit **150***b* starts line-of-sight detection processing on the right eye, and after a line-of-sight detection processing time **146** elapses, the line-of-sight detection unit **150***b* outputs a line-of-sight detection result **148** of the right eye to the control unit **100**. The line-of-sight detection processing time **146** is a time period from when the line-of-sight detection start signal **145** is input to the line-of-sight detection unit **150***b* to when the line-of-sight detection unit **150***b* outputs the line-of-sight detection result **148**. The line-of-sight detection processing includes performing exposure with the eyeball image capturing element **17**, reading out a result of the exposure with the eyeball image capturing element **17**, and performing line-of-sight detection processing using an eyeball image.

(45) The line-of-sight detection units **150***a* and **150***b* have an identical internal configuration to each other, and the line-of-sight detection processing times **136** and **146** are the same time as each other.

(46) Lastly, the control unit **100** can receive the line-of-sight detection result **147** from the line-of-sight detection unit **150***a* and the line-of-sight detection result **148** from the line-of-sight detection unit **150***b* at a cycle of T/2, i.e., 8.3 [ms]. In other words, the line-of-sight detection result **147** from the line-of-sight detection unit **150***a* and the line-of-sight detection result **148** from the line-of-sight detection unit **150***b* are input to the control unit **100** at a cycle of 8.3 [ms]. Thus, the line-of-sight detection results **147** and **148** are input to the control unit **100** at a cycle of 8.3 [ms], providing an improved cycle of reading out the line-of-sight detection results **147** and **148**.

(47) As described above, the HMD **10** includes the line-of-sight detection units **150***a* and **150***b*, and controls the line-of-sight detection start signals **135** and **145** to be transmitted to the line-of-sight detection units **150***a* and **150***b*, respectively, in an optimal manner. With this configuration, the HMD **10** can provide an improved cycle of reading out the line-of-sight detection results **147** and **148**.

(48) Further, in the present exemplary embodiment, the HMD **10** synchronizes the line-of-sight detection units **150***a* and **150***b*, but this does not limit the present disclosure. Even if the background image capturing elements **130***a* and **130***b*, the position detection image capturing elements **140***a* and **140***b*, the display elements **110***a* and **110***b*, or the position-orientation sensor **120** is/are absent, the HMD **10** is applicable. Further, various types of parameter can be designed as appropriate depending on the configuration of the image capturing system **1** and the usage purpose of the HMD **10**

(49) A display vertical synchronization signal to be received from the compositing unit **320** is in synchronization with the background image capturing elements **130***a* and **130***b* and the display elements **110***a* and **110***b*, but may be out of synchronization with the background image capturing elements **130***a* and **130***b* and the display elements **110***a* and **110***b*. In this case, the HMD **10** according to the present exemplary embodiment can perform the above described processing for

improving a cycle of reading out the line-of-sight detection results **147** and **148**.
(50) Each of the background image capturing element **130***b*, the display element **110***b*, and the position detection image capturing element **140***b*, all of which are used for right eyes, may be delayed relative to each of the background image capturing element **130***a*, the display element **110***a*, and the position detection image capturing element **140***a*, all of which are used for left eyes.
(51) There is an MR technique referred to as a seethrough type, by which the HMD **10** with a half mirror arranged in front of both eyes presents a mixed reality created through the half mirror by combining a physical space that a user wearing the HMD **10** is looking at directly and a CG image received from the CG generating unit **310** in the image processing apparatus **30**. With this seethrough type MR technique, the HMD **10** can also perform the above described processing as long as a display vertical synchronization signal, which is to be acquired from the compositing unit **320** as described above, is newly generated using an operation clock in the control unit **100**, and is used as an alternate signal. Even in this case, the HMD **10** can improve a cycle of reading out the line-of-sight detection results **147** and **148**. Further, this case eliminates the need of the display elements **110***a* and **110***b* and the background image capturing elements **130***a* and **130***b*.
(52) As described above, the line-of-sight detection unit **150***a* captures a first eyeball image and detects a first line of sight based on the first eyeball image at the cycle T. The line-of-sight detection unit **150***b* captures a second eyeball image and detects a second line of sight based on the second eyeball image at the cycle T. For example, the first eyeball image is an eyeball image of a left eyeball, and the second eyeball image is an eyeball image of a right eyeball. A timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image are different from each other.
(53) The line-of-sight detection unit **150***a* outputs the line-of-sight detection result **147**. The line-of-sight detection unit **150***b* outputs the line-of-sight detection result **148**. A timing of outputting the line-of-sight detection result **147** and a timing of outputting the line-of-sight detection result **148** are different from each other. The timing of outputting the line-of-sight detection result **148** is different from the timing of outputting the line-of-sight detection result **147** by half the cycle T.
(54) The line-of-sight detection unit **150***a* starts the processing based on the line-of-sight detection start signal **135**. The line-of-sight detection unit **150***b* starts the processing based on the line-of-sight detection start signal **145**. A timing of inputting the line-of-sight detection start signal **135** and a timing of inputting the line-of-sight detection start signal **145** are different from each other. The timing of inputting the line-of-sight detection start signal **145** is different from the timing of inputting the line-of-sight detection start signal **135** by half the cycle T.
(55) The delaying unit **132***b* generates the line-of-sight detection start signal **145** delayed with respect to the line-of-sight detection start signal **135**. The delaying unit **132***a* generates the line-of-sight detection start signal **135** by delaying the display vertical synchronization signal **115***a* by the delay amount D**1**. The delaying unit **132***b* generates the line-of-sight detection start signal **145** by delaying the display vertical synchronization signal **115***b* by the delay amount D**2**.
(56) The display element **110***a* is a display unit that displays images based on the line-of-sight detection result **147**. The display element **110***b* is a display unit that displays images based on the line-of-sight detection result **148**.
(57) Thus, according to the present exemplary embodiment, the HMD **10** can improve a cycle of reading out the line-of-sight detection results **147** and **148** without using an image capturing element supporting high frame rates that causes higher cost or without increasing communication bus width to improve data transfer bandwidth between devices. The HMD **10** controls the timings of transmitting line-of-sight detection start signals **135** and **145** to the line-of-sight detection units **150***a* and **150***b*, respectively, appropriately, providing an improved cycle of reading out the line-of-sight detection start signals **147** and **148**. This enables the HMD **10** to improve a cycle of reading out the line-of-sight detection results **147** and **148**, providing a high-quality line-of-sight detection. This makes it possible for the HMD **10** to perform a rapid detection of lines of sight even if a

composite image moves at high speed.

(58) A second exemplary embodiment will be described. As a technique of measuring the distance to an object a user wearing the HMD **10** is watching, a technique is known of combining left-eye's line-of-sight detection display coordinates, right eye's line-of-sight detection display coordinates, and a focal length of the eyeball image capturing element **17** relative to an origin as a focal point of the eyeball image capturing element **17** and then calculating a vergence angle using trigonometric functions. The farther distance to the object the user wearing the HMD **10** is watching is, the smaller the vergence angle is. Thus, the vergence angle is useful as positional information about the distance to the object. In this case, it is suitable that images obtained from the eyeball image capturing elements **17** in the left eyeball image capturing element **150***a* and the right eyeball image capturing element **150***b* are in synchronization with each other. Thus, to measure a distance to an object the user wearing the HMD **10** is watching, the control unit **100** controls the related units to acquire an image from the eyeball image capturing element **17** in the line-of-sight detection unit **150***a* and an image from the eyeball image capturing element **17** in the line-of-sight detection unit **150***b* in synchronization with each other.

(59) The configuration of the image capturing system **1** according to the second exemplary embodiment is identical to that of the image capturing system **1** according to the first exemplary embodiment illustrated in FIGS. **1**A, **1**B and **2**. The processing performed by the image capturing system **1** according to the second exemplary embodiment is the same as the processing performed by the image capturing system **1** according to the first exemplary embodiment illustrated in FIG. **3**.

(60) FIG. **5** is a time chart illustrating an example of a control operation performed by the HMD **10**, and illustrates timings when images from the eyeball image capturing elements **17** in the line-of-sight detection units **150***a* and **150***b* are acquired, according to the second exemplary embodiment. The following is a description of the difference of FIG. **5** from FIG. **4**.

(61) The control unit **100** receives the display vertical synchronization signal **115** from the compositing unit **320** at a frame cycle of T. The control unit **100** then outputs the display vertical synchronization signal **115***a* to the delaying unit **132***a* and the display vertical synchronization signal **115***b* to the delaying unit **132***b*. It is assumed herein that the display vertical synchronization signals **115***a* and **115***b* are not delayed to each other.

(62) Subsequently, the display vertical synchronization signal **115***a* is input to the delaying unit **132***a*, and then after a period corresponding to a delay amount D**1** elapses, the delaying unit **132***a* outputs the line-of-sight detection start signal **135** to the line-of-sight detection unit **150***a*. The display vertical synchronization signal **115***b* is input to the delaying unit **132***b*, and then after a period corresponding to a delay amount D**2** elapses, the delaying unit **132***b* outputs the line-of-sight detection start signal **145** to the line-of-sight detection unit **150***b*. The delay amount D**1** and the delay amount D**2** are the same as each other. In other words, the timing of outputting the line-of-sight detection start signal **135** and the timing of outputting the line-of-sight detection start signal **145** are the same as each other.

(63) The line-of-sight detection start signal **135** is input to the line-of-sight detection unit **150***a*, and the line-of-sight detection unit **150***a* starts line-of-sight detection processing on the left eye. After the line-of-sight detection processing time **136** elapses, the line-of-sight detection unit **150***a* outputs the line-of-sight detection result **147** of the left eye to the control unit **100**. The line-of-sight detection processing time **136** is a time period from when the line-of-sight detection start signal **135** is input to the line-of-sight detection unit **150***a* to when the line-of-sight detection unit **150***a* outputs the line-of-sight detection result **147**.

(64) The line-of-sight detection start signal **145** is input to the line-of-sight detection unit **150***b*, and the line-of-sight detection unit **150***b* then starts line-of-sight detection processing on the right eye. After the line-of-sight detection processing time **146** elapses, the line-of-sight detection unit **150***b* outputs the line-of-sight detection result **148** of the right eye to the control unit **100**. The line-of-sight detection processing time **146** is a time period from when the line-of-sight detection start

signal **145** is input to the line-of-sight detection unit **150***b* to when the line-of-sight detection unit **150***b* outputs the line-of-sight detection result **148**.
(65) The line-of-sight detection processing times **136** and **146** are the same as each other. The line-of-sight detection result **147** from the line-of-sight detection unit **150***a* and the line-of-sight detection result **148** from the line-of-sight detection unit **150***b* are input to the control unit **100** at the same time. The simultaneous input of the line-of-sight detection results **147** and **148** allows the control unit **100** to measure a distance to an object the user wearing the HMD **10** is watching.
(66) Both of the line-of-sight detection units **150***a* and **150***b* can start line-of-sight detection at a cycle of T. Further, the line-of-sight detection results **147** and **148** from the line-of-sight detection units **150***a* and **150***b* can be input to the control unit **100** at a cycle of T.
(67) As described above, in order that the line-of-sight detection result **147** from line-of-sight detection unit **150***a* and the line-of-sight detection result **148** from the line-of-sight detection unit **150***b* may be input to the control unit **100** at the same time, it is suitable that the delaying units **132***a* and **132***b* output line-of-sight detection start signals **135** and **145**, respectively, at the same time. To perform this processing, it is suitable that the delay amount D**1** to be set at the delaying unit **132***a* and the delay amount D**2** to be set at the delaying unit **132***b* are set to the same value. The display vertical synchronization signal **115** is input to the control unit **100**, and the control unit **100** then transmits the display vertical synchronization signals **115***a* and **115***b* to the delaying units **132***a* and **132***b*, respectively. This allows the control unit **100** to measure the distance to the object the user wearing the HMD **10** is watching.
(68) Similarly, if the control unit **100** measures an interpupillary distance as a distance between the pupils of left and right eyes, it is suitable that the same value is set to the delay amount D**1** at the delaying unit **132***a* and the delay amount D**2** at the delaying unit **132***b*. This allows the control unit **100** to measure the interpupillary distance by acquiring left and right images, in synchronization with each other, from the eyeball image capturing elements **17** in the line-of-sight detection units **150***a* and **150***b*.
(69) Similarly, if line-of-sight calibration on the user wearing the HMD **10** is performed while the control unit **100** displays a standard pattern, it is suitable that the same value is set to the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b*. The control unit **100** can acquire left and right images, in synchronization with each other, from the eyeball image capturing elements **17** in the line-of-sight detection units **150***a* and **150***b*, display the standard pattern, and perform the line-of-sight calibration on the user wearing the HMD **10**.
(70) On the other hand, if a low priority is given to the calculation of the distance to the object due to the increased distance to the object the user wearing the HMD **10** is watching, the control unit can detect that fact, and make the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* different from each other. This allows the operations performed by the line-of-sight detection units **150***a* and **150***b* to be changed dynamically. For example, the moment a menu operation graphical user interface (GUI) is displayed, the control units detects that fact, and sets the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* to different values from each other, to enhance the performance of acquiring line-of-sight detection results. This is applicable to cases, such as a case of improving a cycle of reading out a line-of-sight detection result.
(71) Another example is a case where a virtual reality (VR) image composed of a CG image is to be displayed. In this case, the control unit **100** can detect that fact, and set the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* to different values from each other, to enhance the performance of acquiring line-of-sight detection results.
(72) Thus, a cycle of reading out a line-of-sight detection result may be improved.
(73) Further, yet another example is a case where the movement of an object being watched is slow. In this case, the control unit **100** can detect no change in the movement of the object, and set the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* to different values from each other, to enhance the performance of acquiring line-of-sight detection results. This improves a cycle of

reading out a line-of-sight detection result.

(74) Further, yet another example is a case of a line-of-sight tracking priority mode, such as foveated rendering. In this case, the control unit **100** can set the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* to different values from each other, to improve a cycle of reading out a line-of-sight detection result. This improves the speed of acquiring line-of-sight detection results.

(75) As yet another example, the control unit **100** calculates an average of display coordinates of left eye's line-of-sight detection results and an average of display coordinates of right eye's line-of-sight detection results. After that, if the difference between the average of display coordinates of left eye's line-of-sight detection results and the average of display coordinates of right eye's line-of-sight detection results exceeds a threshold, the control unit **100** sets the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* to different values from each other. This improves the speed of acquiring line-of-sight detection results. On the other hand, if the difference between the average of display coordinates of left eye's line-of-sight detection results and the average of display coordinates of right eye's line-of-sight detection results falls within the threshold, the control unit **100** sets the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* to the same value as each other. This enhances the reliability of line-of-sight detection by acquiring left and right images, in synchronization with each other, from the eyeball image capturing elements **17** in the line-of-sight detection units **150***a* and **150***b*. Consequently, the control unit **100** can dynamically change both values using a threshold.

(76) As described above, the line-of-sight detection unit **150***a* captures a first eyeball image and detects a first line of sight based on the first eyeball image at a cycle of T. The line-of-sight detection unit **150***b* captures a second eyeball image and detects a second line of sight based on the second eyeball image at a cycle of T. For example, the first eyeball image is a left eyeball image, and the second eyeball image is a right eyeball image. The difference between a timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image is changed depending on the state of the HMD **10**, the user setting, and the operation from the user.

(77) The control unit **100** can calculate a parallax, a vergence angle, or an interpupillary distance or perform line-of-sight calibration based on the line-of-sight detection result **147** and the line-of-sight detection result **148**. In this case, the control unit **100** controls a timing of starting to capture the first eyeball image by the line-of-sight detection unit **150***a* and a timing of starting to capture the second eyeball image by the line-of-sight detection unit **150***b* to be the same as each other.

(78) The display element **110***a* displays an image based on the line-of-sight detection result **147**. The display element **110***b* displays an image based on the line-of-sight detection result **148**. When the display elements **110***a* and **110***b* display operation screens through user operation, the control unit **100** controls a timing of starting to capture the first eyeball image by the line-of-sight detection unit **150***a* and a timing of starting to capture the second eyeball image by the line-of-sight detection unit **150***b* to be the different from each other.

(79) The control unit **100** calculates a vergence angle based on the line-of-sight detection results **147** and **148**. If the vergence angle is smaller than a threshold, the control unit **100** controls a timing of starting to capture the first eyeball image by the line-of-sight detection unit **150***a* and a timing of starting to capture the second eyeball image by the line-of-sight detection unit **150***b* to be different from each other.

(80) Consequently, according to the present exemplary embodiment, the control unit **100** can change the performance of acquiring line-of-sight detection results with flexibility by changing the delay amounts D**1** and D**2** at the delaying units **132***a* and **132***b* dynamically, providing line-of-sight detection functions responding to requests from various user applications.

Other Embodiments

(81) Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-

transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc™ (BD)), a flash memory device, a memory card, and the like.
(82) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(83) This application claims the benefit of Japanese Patent Application No. 2022-116420, filed Jul. 21, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image capturing apparatus comprising: a first line-of-sight detector configured to capture a first eyeball image as a left eyeball image and detect a first line of sight based on the first eyeball image at a first cycle; a second line-of-sight detector configured to capture a second eyeball image as a right eyeball image and detect a second line of sight based on the second eyeball image at the first cycle; a first display unit configured to display a first image based on a result of detecting the first line of sight; and a second display unit configured to display a second image based on a result of detecting the second line of sight, wherein a timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image are different from each other.
2. The image capturing apparatus according to claim 1, wherein the first line-of-sight detector outputs a result of detecting the first line of sight, wherein the second line-of-sight detector outputs a result of detecting the second line of sight, and wherein a timing of outputting the result of detecting the first line of sight and a timing of outputting the result of detecting the second line of sight are different from each other.
3. The image capturing apparatus according to claim 1, wherein the first line-of-sight detector starts processing based on a first start signal, wherein the second line-of-sight detector starts processing based on a second start signal, and wherein a timing of inputting the first start signal and a timing of inputting the second start signal are different from each other.
4. The image capturing apparatus according to claim 1, wherein the timing of starting to capture the second eyeball image is different from the timing of starting to capture the first eyeball image by a period corresponding to a half of the first cycle.
5. An image capturing apparatus comprising: a first line-of-sight detector configured to capture a first eyeball image as a left eyeball image and detect a first line of sight based on the first eyeball image at a first cycle and output a result of detecting the first line of sight; and a second line-of-sight detector configured to capture a second eyeball image as a right eyeball image and detect a second line of sight based on the second eyeball image at the first cycle and output a result of detecting the second line of sight, wherein the timing of outputting the result of detecting the

second line of sight is different from the timing of outputting the result of detecting the first line of sight by a period corresponding to a half of the first cycle.

6. The image capturing apparatus according to claim 3, wherein the timing of inputting the second start signal is different from the timing of inputting the first start signal by a period corresponding to a half of the first cycle.

7. The image capturing apparatus according to claim 3, further comprising a delaying unit configured to generate the second start signal that is delayed with respect to the first start signal.

8. The image capturing apparatus according to claim 7, wherein the delaying unit generates the first start signal by delaying a display vertical synchronization signal by a first delay amount, and generates the second start signal by delaying the display vertical synchronization signal by a second delay amount.

9. An image capturing apparatus comprising: a first line-of-sight detector configured to capture a first eyeball image as a left eyeball image and detect a first line of sight based on the first eyeball image at a first cycle; and a second line-of-sight detector configured to capture a second eyeball image as a right eyeball image and detect a second line of sight based on the second eyeball image at the first cycle, wherein a difference between a timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image is changed depending on a state of the image capturing apparatus, a user setting, or an operation from a user.

10. The image capturing apparatus according to claim 9, further comprising a control unit configured to control, in a case where a parallax, a vergence angle, or an interpupillary distance is calculated, or line-of-sight calibration is performed, based on a result of detecting the first line of sight and a result of detecting the second line of sight, a timing of starting to capture the first eyeball image by the first line-of-sight detector and a timing of starting to capture the second eyeball image by the second line-of-sight detector to be the same as each other.

11. The image capturing apparatus according to claim 9, further comprising: a first display unit configured to display a first image based on a result of detecting the first line of sight; and a second display unit configured to display a second image based on a result of detecting the second line of sight.

12. The image capturing apparatus according to claim 11, further comprising a control unit configured to control, in a case where operation screens are displayed on the first display unit and the second display unit by operation performed by a user, the timing of starting to capture the first eyeball image by the first line-of-sight detector and the timing of starting to capture the second eyeball image by the second line-of-sight detector to be different from each other.

13. The image capturing apparatus according to claim 9, further comprising a control unit configured to control, in a case where a vergence angle calculated based on a result of detecting the first line of sight and a result of detecting the second line of sight is smaller than a first threshold, the timing of starting to capture the first eyeball image by the first line-of-sight detector and the timing of starting to capture the second eyeball image by the second line-of-sight detector to be different from each other.

14. An image capturing method for an image capturing apparatus, the image capturing method comprising: capturing a first eyeball image as a left eyeball image and detecting a first line of sight based on the first eyeball image at a first cycle; capturing a second eyeball image as a right eyeball image and detecting a second line of sight based on the second eyeball image at the first cycle; displaying, on a first display unit, a first image based on a result of detecting the first line of sight; and displaying, on a second display unit, a second image based on a result of detecting the second line of sight, wherein a timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image are different from each other.

15. A non-transitory computer-readable storage medium storing a program that causes a computer to execute an image capturing method for an image capturing apparatus, the image capturing method comprising: capturing a first eyeball image as a left eyeball image and detecting a first line

of sight based on the first eyeball image at a first cycle; capturing a second eyeball image as a right eyeball image and detecting a second line of sight based on the second eyeball image at the first cycle; displaying, on a first display unit, a first image based on a result of detecting the first line of sight; and displaying, on a second display unit, a second image based on a result of detecting the second line of sight, wherein a timing of starting to capture the first eyeball image and a timing of starting to capture the second eyeball image are different from each other.